PRODUCT DATA EXPLORATION USING EXCEL

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1. Introduction

This document provides a complete analysis of a product dataset using Microsoft Excel. It includes summary calculations, logical functions, conditional formulas, and text extraction techniques to understand pricing patterns and category-wise distribution.

2. Purpose

The purpose of this analysis is to explore a product dataset using Excel and apply essential formulas such as SUM, COUNT, AVERAGE, IF, SUMIF, COUNTIF, LEFT, and RIGHT. This helps in understanding product pricing, category distribution, and extracting meaningful insights from raw data.

3. Combined Output Values and Formulas (Side by Side)

3.1 Sum, Count, and Average

- **Total Price of All Products:** ₹10,100.00

  Formula: =SUM(D3:D36)

- **Total Number of Products:** 34

  Formula: =COUNT(E3:E36)

- **Average Price of Products:** ₹297.06

  Formula: =AVERAGE(D3:D36)

3.2 Minimum and Maximum Price

- **Minimum Price:** ₹30.00

  Formula: =MIN(D3:D36)

- **Maximum Price:** ₹1,000.00

  Formula: =MAX(D3:D36)

3.3 Logical Function

- **High Price / Standard Price Classification**

  Formula: =IF(D3>500,"High Price","Standard Price")

3.4 Conditional Functions

- **Total Price of Electronics Category:** ₹8,050.00

  Formula: =SUMIF(F3:F36,"Electronics",D3:D36)

- **Number of Products Below ₹100:** 11

  Formula: =COUNTIF(D3:D36,"<100")

3.5 Text Extraction

- **Extract Day: Formula:** =LEFT(A3,2)
- **Extract Country Code: Formula:** =RIGHT(A3,2)
- **Extract Month: Formula:** =MID(A3,4,3)

4. Key Insights

- Electronics category contributes the highest total value.
- 11 products fall under the low-price segment (< ₹100).
- Logical formula helps identify premium vs. standard products.
- Text functions help decode information from Product ID.

5. Conclusion

Excel formulas such as SUM, COUNT, AVERAGE, IF, SUMIF, COUNTIF, LEFT, and RIGHT help extract meaningful insights from product datasets. This analysis provides a clear understanding of pricing patterns and category distribution, useful for business and academic reporting.

6. Prepared By

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